

MUMBA AMOS NTAMBO

Full Stack Developer



PERSONAL DATA

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Date of Birth: 29.04.2002 | **Nationality:** Zambian

EXPERIENCE

AI / Full-Stack Engineer (Freelance) | Baseline Health AI Platform, *Kuala Lumpur, Malaysia* **June 2025 to May 2026**
Healthcare AI start-up: Patient analytics and reporting platform
[baseline.website](#)

- Built a women's health platform end to end: Next.js patient and clinic portals with a FastAPI service and Supabase Postgres, deployed on Vercel.
 - Used GPT-4o vision to extract structured biomarker data from scanned lab reports, replacing manual entry of hormone panels and cutting report turnaround time by around 80%.
 - Generated clinical narratives with GPT-5 using structured outputs against a strict JSON schema, with all statistics computed in code so the model narrates figures rather than producing them.
 - Built a two-store RAG layer on pgvector (HNSW cosine, text-embedding-3-small) giving each generated report patient-scoped retrieval over prior analyses and medical documents.
 - Measured cross-source similarity as non-discriminating (0.043 vs 0.157 score spread) and switched that path to recency selection, cutting an embedding call per generation.
 - Made retrieval fail-soft so embedding or RPC errors degrade output instead of blocking generation.
 - Built the access-control layer for a multi-tenant clinical platform: bearer-token identity resolved server-side on every request, with patient-facing routes deriving the patient from the token rather than any client-supplied ID.
 - Scoped clinic staff to their own clinic's patients through a shared authorization module with read-only enforcement, and served medical files exclusively through one-hour signed URLs from private buckets, never public URLs.
 - Designed data handling to meet Malaysia's PDPA obligations for sensitive personal data: purpose-limited collection, per-clinic access scoping, encrypted storage, and medical records reachable only through short-lived signed access.
- Tech:** Python, FastAPI, Next.js, OpenAI GPT-4o/GPT-5, pgvector, Supabase, Vercel, PostgreSQL

DevOps / Platform Engineer (Freelance) | GoAcara Event Management Platform, *Kuching, Malaysia* **August 2025 to January 2026**
Microservices-based event management SaaS
[goacara.com](#)

- Provisioned a 4-node EKS cluster on AWS with cluster auto-scaling and ALB ingress, running a microservices event platform in production.
 - Wrote the full infrastructure in Terraform with remote state in S3, so the environment could be torn down and rebuilt from code rather than reconstructed by hand.
 - Built GitOps delivery with ArgoCD and GitHub Actions, so deployments came from committed manifests rather than manual kubectl.
 - Containerised services with Podman and Docker to keep local development and production builds identical, removing environment drift as a class of bug.
 - Ran the platform end to end as sole engineer: cluster, pipelines, database, and application code.
- Tech:** Vue.js, Nuxt.js, Node.js, Go, PostgreSQL, Docker, Podman, Kubernetes, AWS EKS, ArgoCD, Terraform, GitHub Actions

Software Engineering Intern | Sarawak Research Development Council, *Kuching, Malaysia* **January 2024 to December 2024**
Malaysian state government research agency
[rpmn.sarawakrdc.org.my](#) (live internal system, login required)

- Collaborated with a 5-person team to build a grant application management system with React and TypeScript frontend, and a Node.js (Express) backend.
 - Built the initial backend of the system in Laravel (PHP) before the project moved to Node.js (Express).
 - Enabled real-time tracking of researcher profiles, publications, and performance metrics across departments.
 - Deployed on DigitalOcean with Linux shell scripting and cron jobs for automated processing and backup.
- Tech:** React, TypeScript, Node.js, Express, Laravel, PHP, EJS, SQL, DigitalOcean, Linux

IT Project Intern | Malaysian Dutch Business Council, *Kuala Lumpur, Malaysia* **September 2021 to December 2021**
Bilateral trade association supporting Dutch and Malaysian businesses

- Built an invoicing feature inside the organisation's Bitrix24 CRM, working within an existing CRM/ERP system rather than a greenfield codebase.
 - Worked with CRM data structures, records, and workflow configuration to fit the tool to how the team actually processed members and billing.
- Tech:** Bitrix24 CRM, PHP, SQL, HTML/CSS

EDUCATION

BSc Computer Science in Artificial Intelligence | Swinburne University of Technology, Sarawak Campus, Malaysia **2022 to 2025**
Australian Bachelor's degree (AQF Level 7), awarded by Swinburne Melbourne

- Relevant coursework: Machine Learning, Deep Learning, Data Structures and Algorithms, Software Engineering, Cloud Computing, Database Systems.
- Activities: Vice President, Africa Unites Society (Swinburne).

Diploma in Business Information Technology | Asia Pacific University (APU), Kuala Lumpur, Malaysia **2019 to 2021**
MQA-accredited private university in Malaysia

PERSONAL PROJECTS

Trainioapp | Personal Trainer and Client Marketplace App **August 2025 to March 2026**

Cross-platform mobile app, live on the App Store and Google Play
[App Store](#) | [Google Play](#)

- Built a full-stack cross-platform mobile app connecting clients with personal trainers, featuring swipe-based trainer discovery, real-time chat, and session booking.
- Implemented role-based flows for trainers and clients including profile management, certifications, earnings tracking, client leads, and workout/diet plan builder.
- Integrated in-app subscriptions with RevenueCat and gated Pro features; published builds to Google Play and Apple App Store.

Tech: React Native, Expo, TypeScript, Supabase (PostgreSQL + Auth), RevenueCat, Google Maps API

PupMood | Dog Mood Tracking Mobile App

June 2026

Cross-platform mobile app, live on the App Store
[App Store](#)

- Built a full-stack cross-platform mobile app for logging and analyzing a dog's mood over time, featuring multi-dog support, streak tracking, time-of-day mood bucketing, and animated UI with haptic and audio feedback.
- Integrated Supabase for backend, authentication, and data sync, with Google and Apple Sign-In and a route-guarded onboarding flow.
- Implemented in-app subscriptions with RevenueCat, gating multi-dog support, mood trends, and AI features behind entitlements.
- Added AI-powered mood summaries that turn logged entries into natural-language insights, plus push notifications and reminders.
- Handled the full release pipeline: EAS Build, App Store submission, and a marketing landing page with legal/support pages.

Tech: React Native, Expo, TypeScript, Supabase, RevenueCat, EAS Build

TECHNICAL SKILLS

Languages and Frameworks: Python, TypeScript, JavaScript, PHP, Go, SQL, HTML/CSS, FastAPI, Laravel, React.js, Next.js, Vue.js, Nuxt.js, Node.js, Express, React Native

Cloud and DevOps: AWS (EKS, S3, ALB), Kubernetes, Docker, Podman, Terraform, ArgoCD, GitHub Actions, CI/CD, Linux

Databases: PostgreSQL, SQL, Supabase, DigitalOcean managed DBs

AI and Machine Learning: RAG, vector search (pgvector), embeddings, prompt engineering, LLM integration, TensorFlow, OpenAI GPT-4o/GPT-5 API, KNIME Analytics

CERTIFICATIONS

Machine Learning Specialization, DeepLearning.AI (June 2025)
Kubernetes Hands-On: Deploy Microservices to AWS Cloud, Udemy (2025)

LANGUAGES

English: Native | German: A1 (in progress) | Dutch: Beginner

ACHIEVEMENTS & ACTIVITIES

Vice President, Africa Unites Society, Swinburne University Sarawak (2023 to 2024).